AWS VPC PEERING

PRIVATE CONNECTIVITY PROJECT REPORT

Connect two Amazon VPCs and validate private communication

Cover details	Enter your details
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Batch / Course	Batch 1/Devops
Project title	AWS VPC Peering – Private Connectivity
AWS Region	Asia Pacific (Sydney) – ap-southeast-2
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1. Executive Summary

This project demonstrates private network connectivity between two Amazon Virtual Private Clouds (VPCs) using a VPC peering connection. VPC A contains a public bastion host and a private application instance (App-A). VPC B contains a private application instance (App-B). The application instances communicate using private IPv4 addresses over the peering connection.

A bastion host in VPC A provides an administrative entry point. App-A and App-B are intended to remain private with no public IPv4 addresses. Custom route tables provide routes in both directions, while security groups restrict inbound SSH and ICMP traffic.

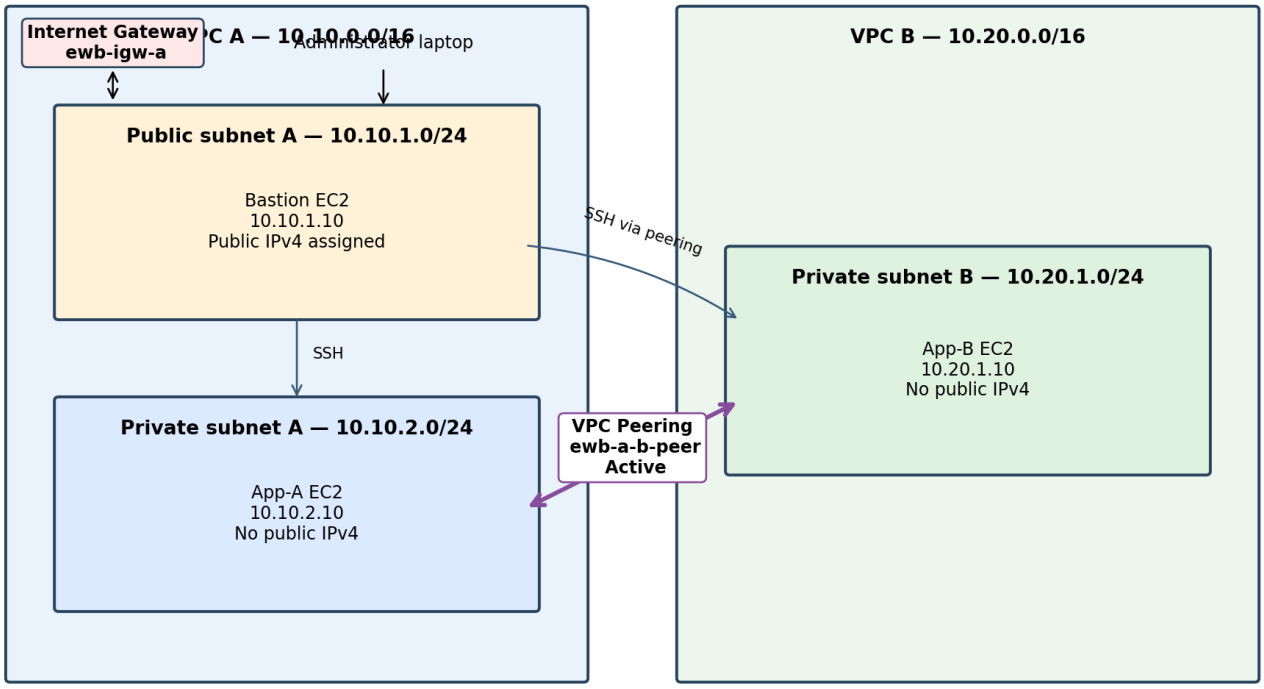
The reported validation included successful ICMP tests in both directions, SSH access to App-B through the bastion, and a controlled ICMP rule-removal test followed by recovery.

2. Project Objectives

- Plan non-overlapping VPC and subnet CIDRs.
- Create public and private subnets with explicit route-table associations.
- Launch one bastion host and two private EC2 instances.
- Activate VPC peering and configure forward and return routes.
- Apply least-privilege security-group rules for SSH and ICMP.
- Verify private connectivity and demonstrate controlled failure and recovery.
- Document the architecture, test results and cleanup.

3. Architecture Diagram

The diagram represents the project design. Verify the final console configuration and fill in actual AWS resource IDs in the inventory.



Private workloads communicate using private IPv4 addresses; peering is non-transitive.

App-A and App-B exchange traffic using private IP addresses over VPC peering. SSH administration to App-B is performed through the bastion host in VPC A.

4. AWS Resource Inventory

Resource	Name	CIDR / private IP / notes
Region	Sydney	ap-southeast-2
VPC A	ewb-vpc-a	10.10.0.0/16
Public subnet A	ewb-public-subnet-a	10.10.1.0/24
Private subnet A	ewb-private-subnet-a	10.10.2.0/24
VPC B	ewb-vpc-b	10.20.0.0/16
Private subnet B	ewb-private-subnet-b	10.20.1.0/24
Internet Gateway	ewb-igw-a	Attached to VPC A
VPC peering	ewb-a-b-peer	Active
Bastion EC2	ewb-bastion	10.10.1.10; public IPv4 assigned
App-A EC2	ewb-app-a	10.10.2.10; no public IPv4
App-B EC2	ewb-app-b	10.20.1.10; no public IPv4
Route tables	ewb-rt-a-public / ewb-rt-a-private / ewb-rt-b-private	See Section 6
Security groups	ewb-sg-bastion / ewb-sg-app-a / ewb-app-b-sg	App-B SG name as observed

Resource ID field	Copy actual ID from AWS Console
VPC A ID	vpc-04c026bfb392e799a
VPC B ID	vpc-05ced7aaf20131816
Public subnet A ID	subnet-0ee8bc8b8d2d48748
Private subnet A ID	subnet-01c9998d5487df3c6
Private subnet B ID	subnet-0798f695629b74c60
Peering connection ID (pcx-...)	pcx-07263439d3860b5e5
Bastion instance ID	i-0324e2b670e6d55a2
App-A instance ID	i-0015475ea6e3bfdf3

App-B instance ID	i-00d1c985c4b5c3ef6
Public route table ID	subnet-0ee8bc8b8d2d48748
Private A route table ID	subnet-01c9998d5487df3c6
Private B route table ID	subnet-0798f695629b74c60

5. Network and Subnet Plan

VPC A uses 10.10.0.0/16 and VPC B uses 10.20.0.0/16. These ranges do not overlap, allowing the peering connection to route traffic between the VPCs.

VPC	Subnet	CIDR	Purpose	Route table
VPC A	ewb-public-subnet-a	10.10.1.0/24	Bastion	ewb-rt-a-public
VPC A	ewb-private-subnet-a	10.10.2.0/24	App-A	ewb-rt-a-private
VPC B	ewb-private-subnet-b	10.20.1.0/24	App-B	ewb-rt-b-private

Figure 1 — VPC and subnet configuration

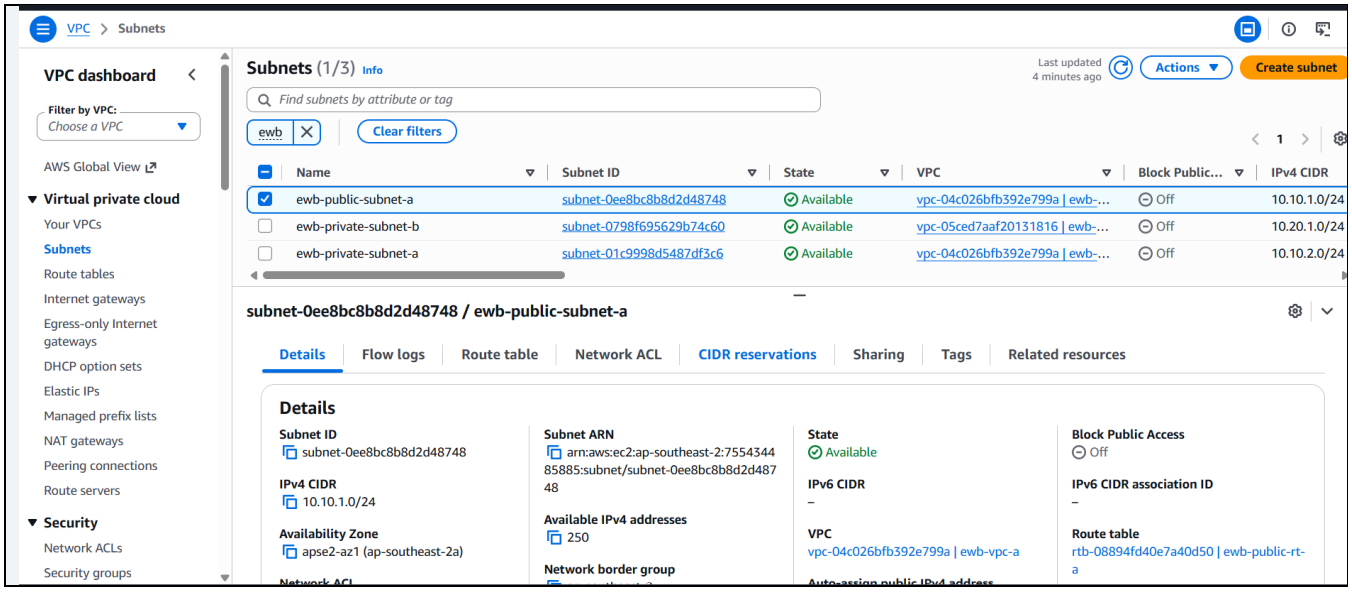
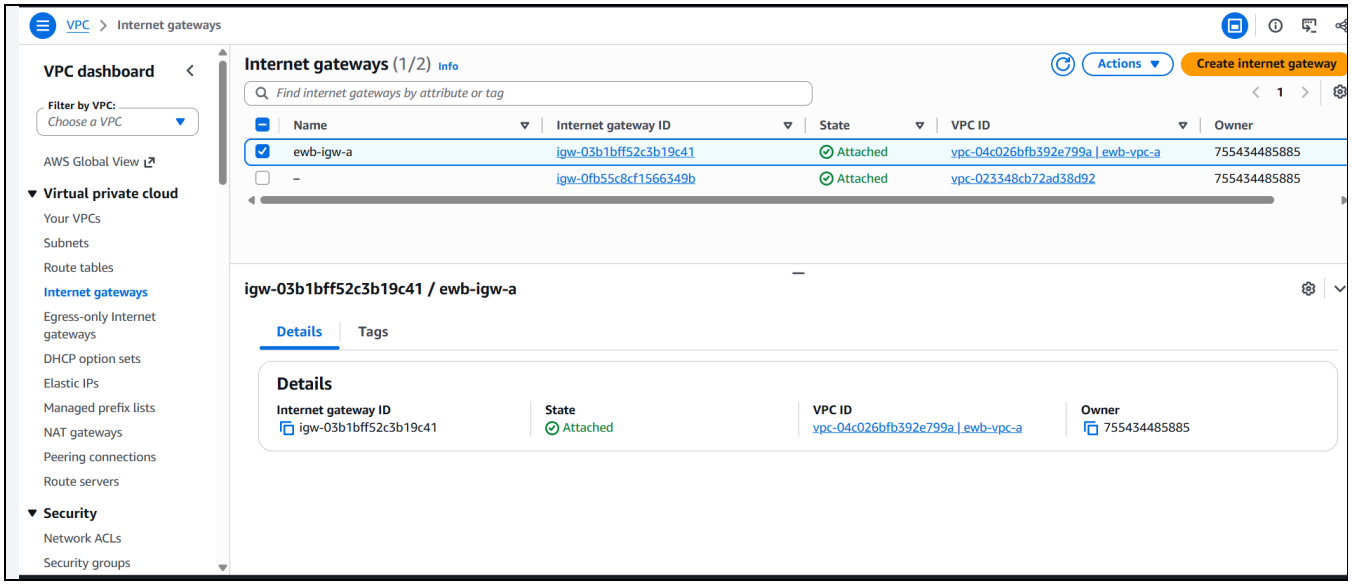


Figure 2 — Internet Gateway



6. VPC Peering and Route Tables

The peering connection ewb-a-b-peer links ewb-vpc-a and ewb-vpc-b. Its status was verified as Active. Peering is non-transitive: it connects the two peer VPCs only.

Route table	Subnet association	Destination	Target
ewb-rt-a-public	ewb-public-subnet-a	10.10.0.0/16	local
ewb-rt-a-public	ewb-public-subnet-a	0.0.0.0/0	Internet Gateway ewb-igw-a
ewb-rt-a-public	ewb-public-subnet-a	10.20.0.0/16	Peering ewb-a-b-peer
ewb-rt-a-private	ewb-private-subnet-a	10.10.0.0/16	local
ewb-rt-a-private	ewb-private-subnet-a	10.20.0.0/16	Peering ewb-a-b-peer
ewb-rt-b-private	ewb-private-subnet-b	10.20.0.0/16	local
ewb-rt-b-private	ewb-private-subnet-b	10.10.0.0/16	Peering ewb-a-b-peer

Both directions need routes. App-A traffic uses the VPC A private route table to reach 10.20.0.0/16 through peering. Return traffic uses the VPC B private route table to reach 10.10.0.0/16 through the same peering connection. The public route table also has a route to VPC B so the bastion can reach App-B.

Figure 3 — Active peering connection

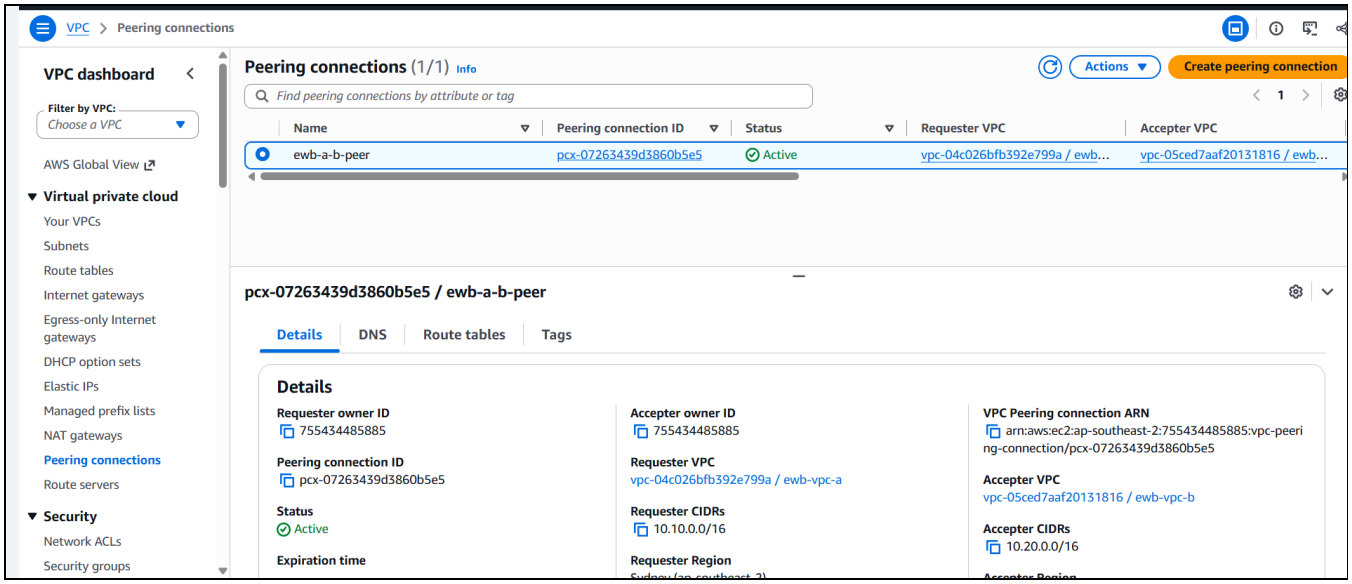


Figure 4 — Public route table and association

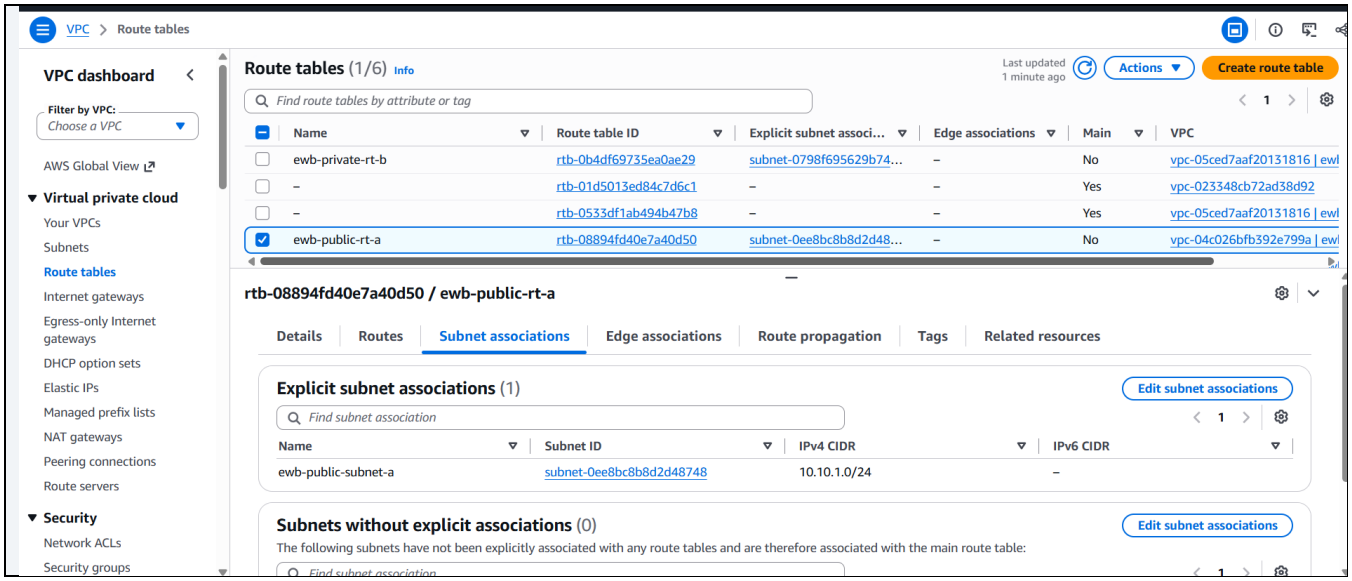


Figure 5 — Private route table A and association

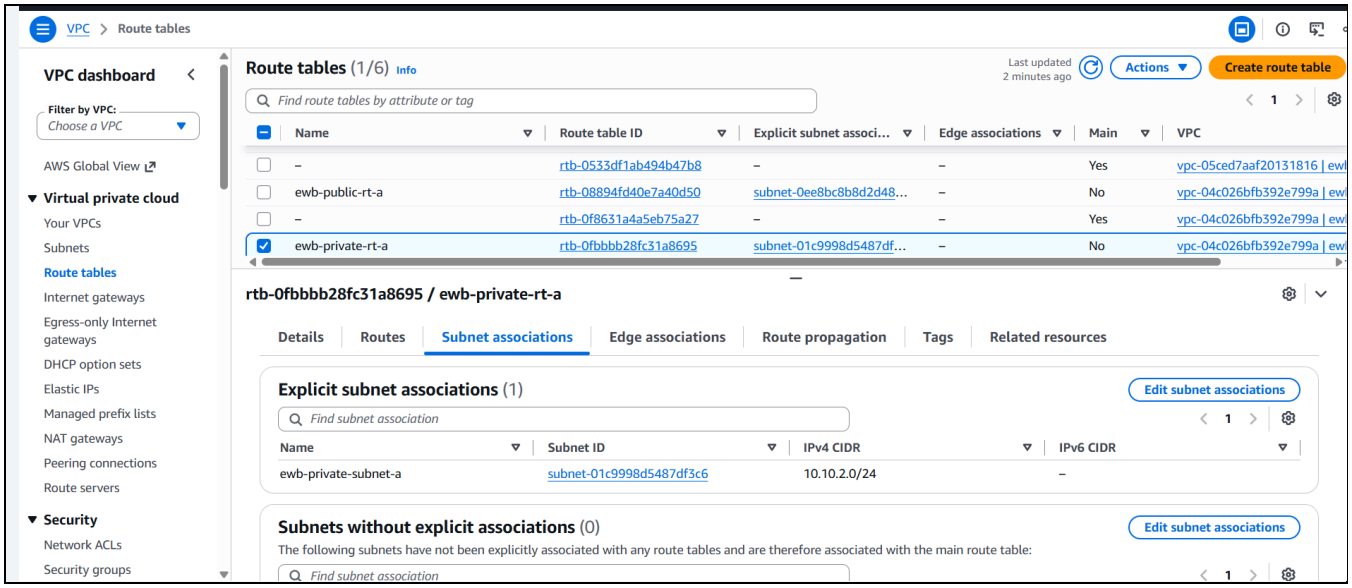
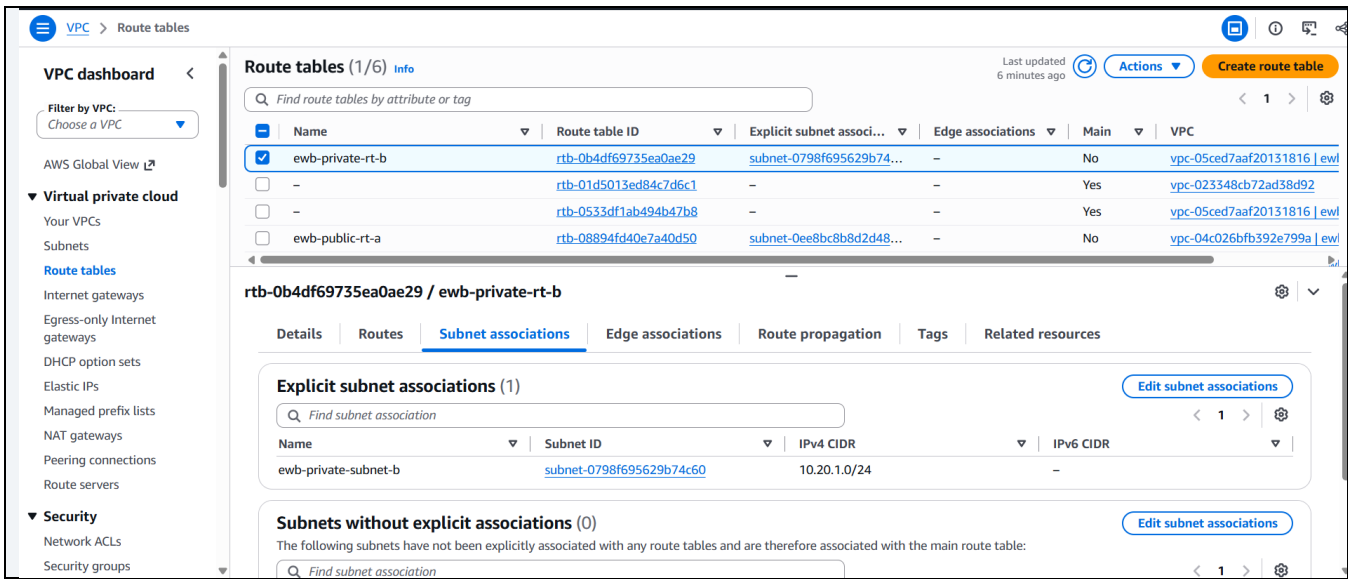


Figure 6 — Private route table B and association



7. EC2 Instances and Security Groups

Instance	Subnet	Private IPv4	Public IPv4
ewb-bastion	Public subnet A	10.10.1.10	Assigned
ewb-app-a	Private subnet A	10.10.2.10	None
ewb-app-b	Private subnet B	10.20.1.10	None

The application instances are private workloads. App-B was accessed from the administrator's laptop through the bastion using SSH forwarding (ProxyCommand). The private key must remain private and must not be included in the report.

7.1 Intended inbound security rules

Security group	Inbound rule	Allowed source
ewb-sg-bastion	SSH / TCP 22	Administrator public IPv4 /32
ewb-sg-app-a	SSH / TCP 22	10.10.1.10/32 (Bastion)

ewb-sg-app-a	ICMP Echo Request	10.20.1.10/32 (App-B)
ewb-app-b-sg	SSH / TCP 22	10.10.1.10/32 (Bastion)
ewb-app-b-sg	ICMP Echo Request	10.10.2.10/32 (App-A)

These are the intended least-privilege rules. Compare them with the final console rules before submission. Confirm ICMP is restricted to Echo Request if required, and remove temporary troubleshooting rules. Do not expose private-instance SSH to 0.0.0.0/0.

Figure 7 — Bastion security group

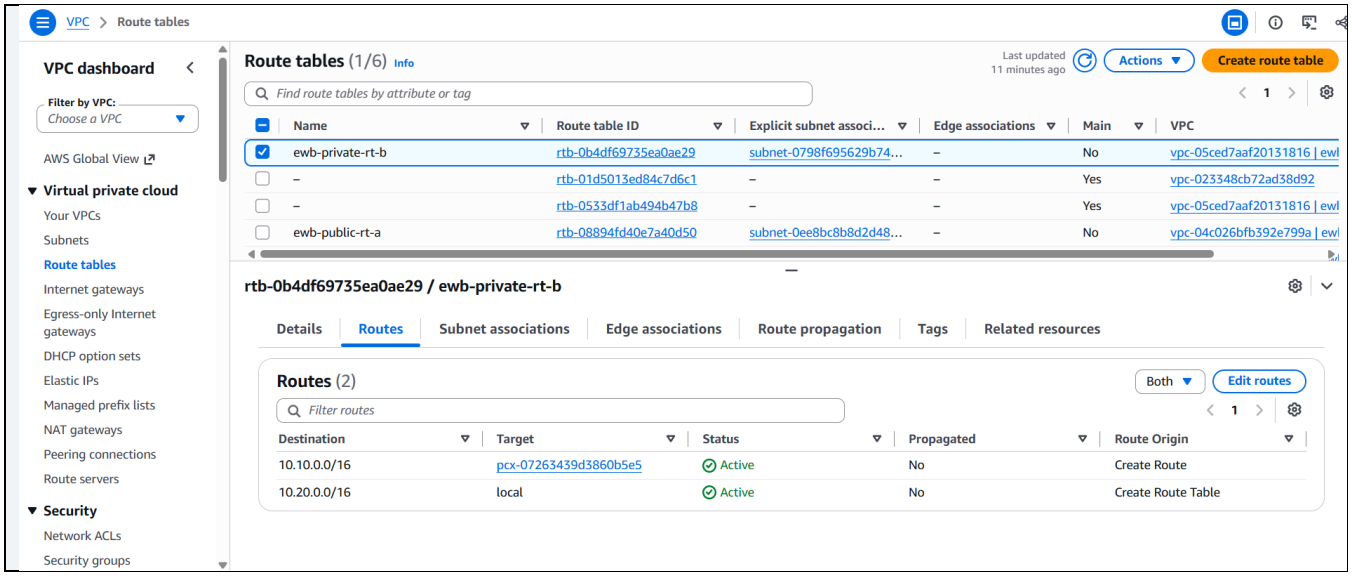


Figure 8 — App-A security group

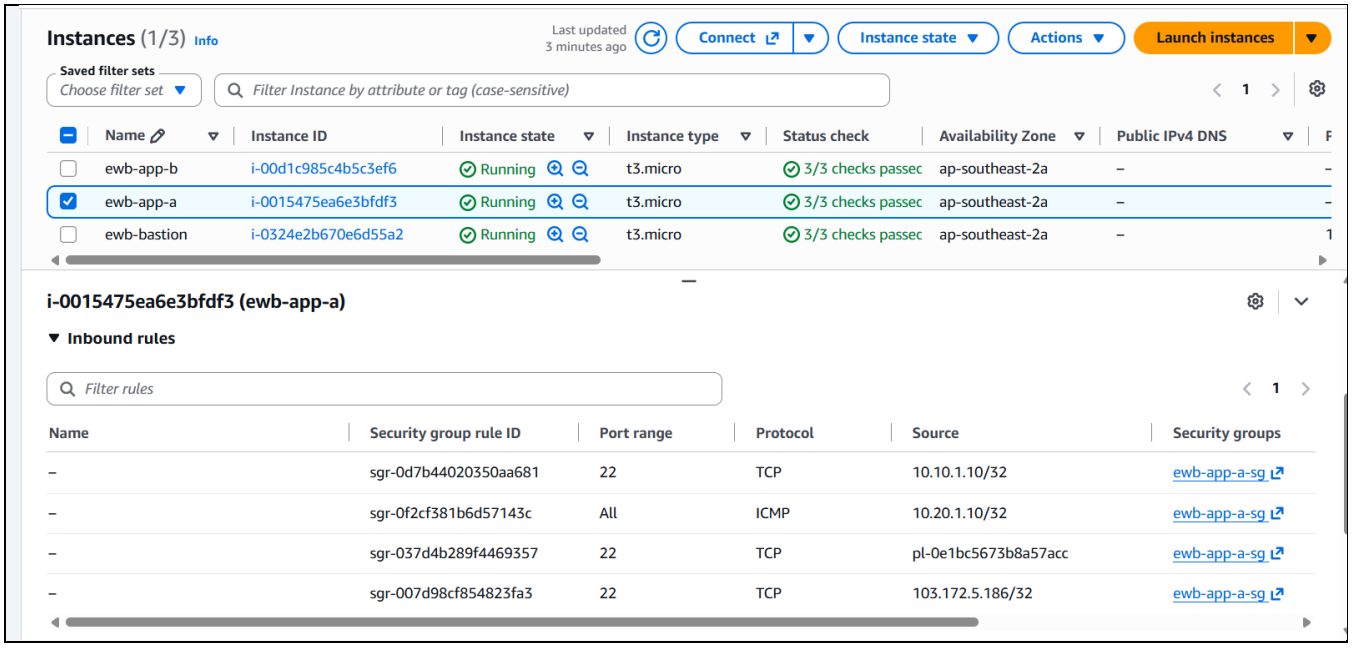


Figure 9 — App-B security group

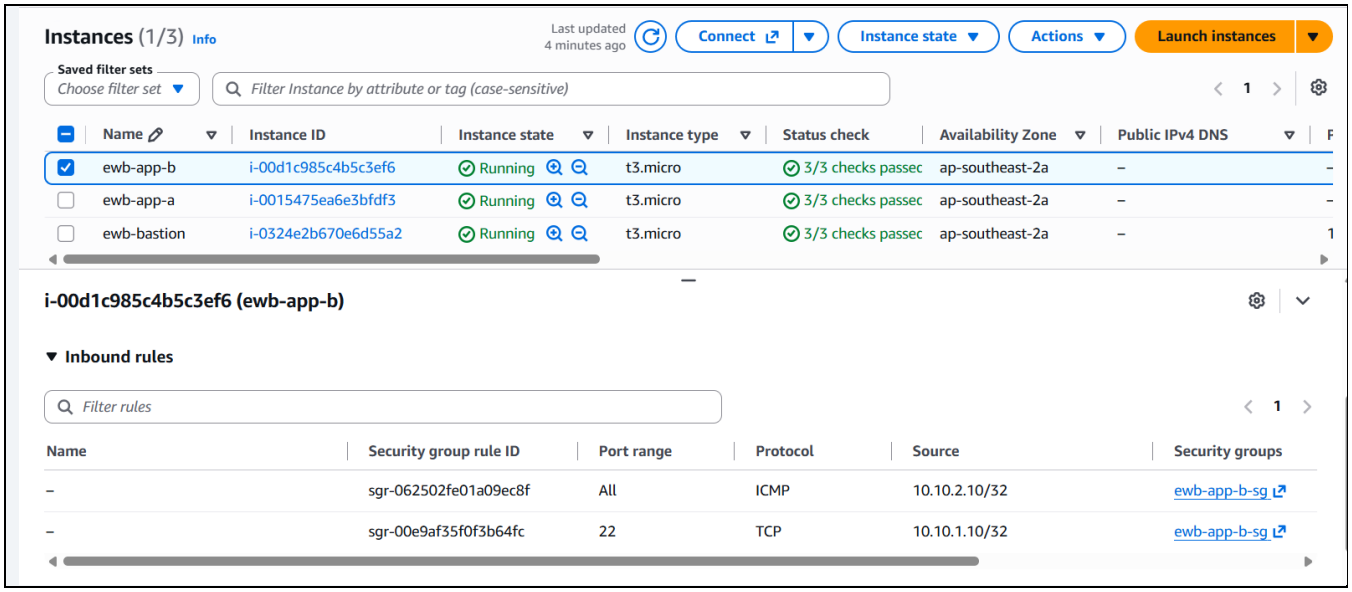


Figure 10 — EC2 instances

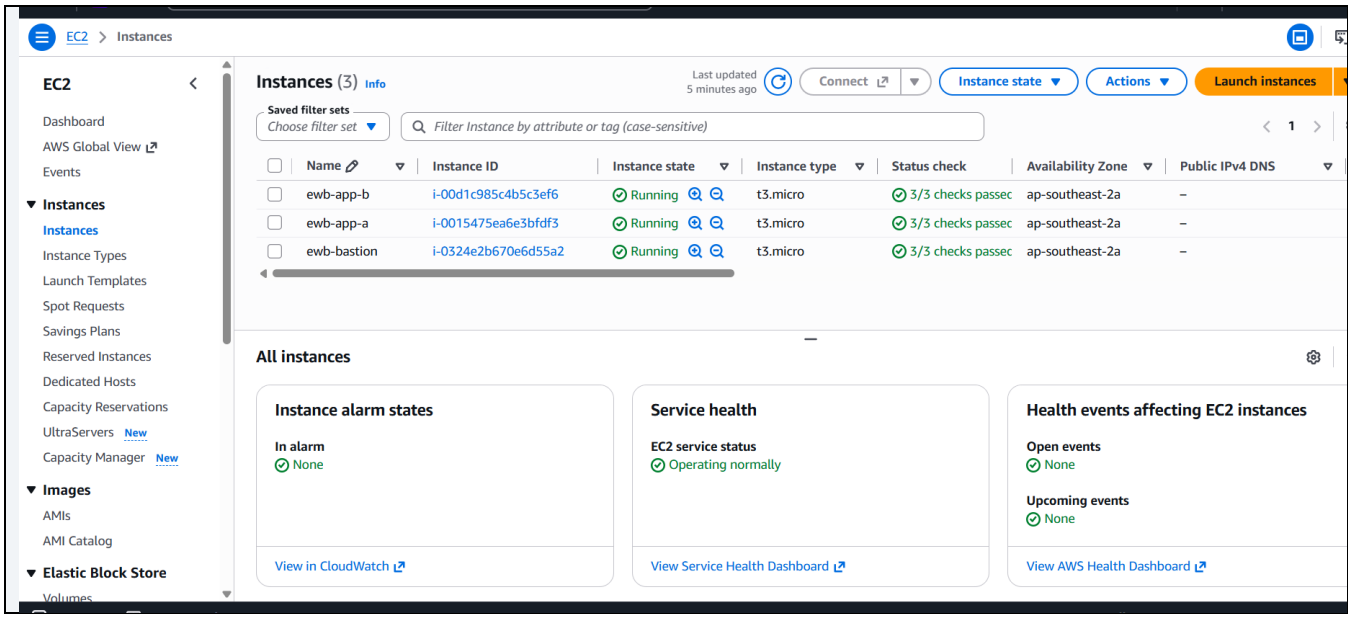
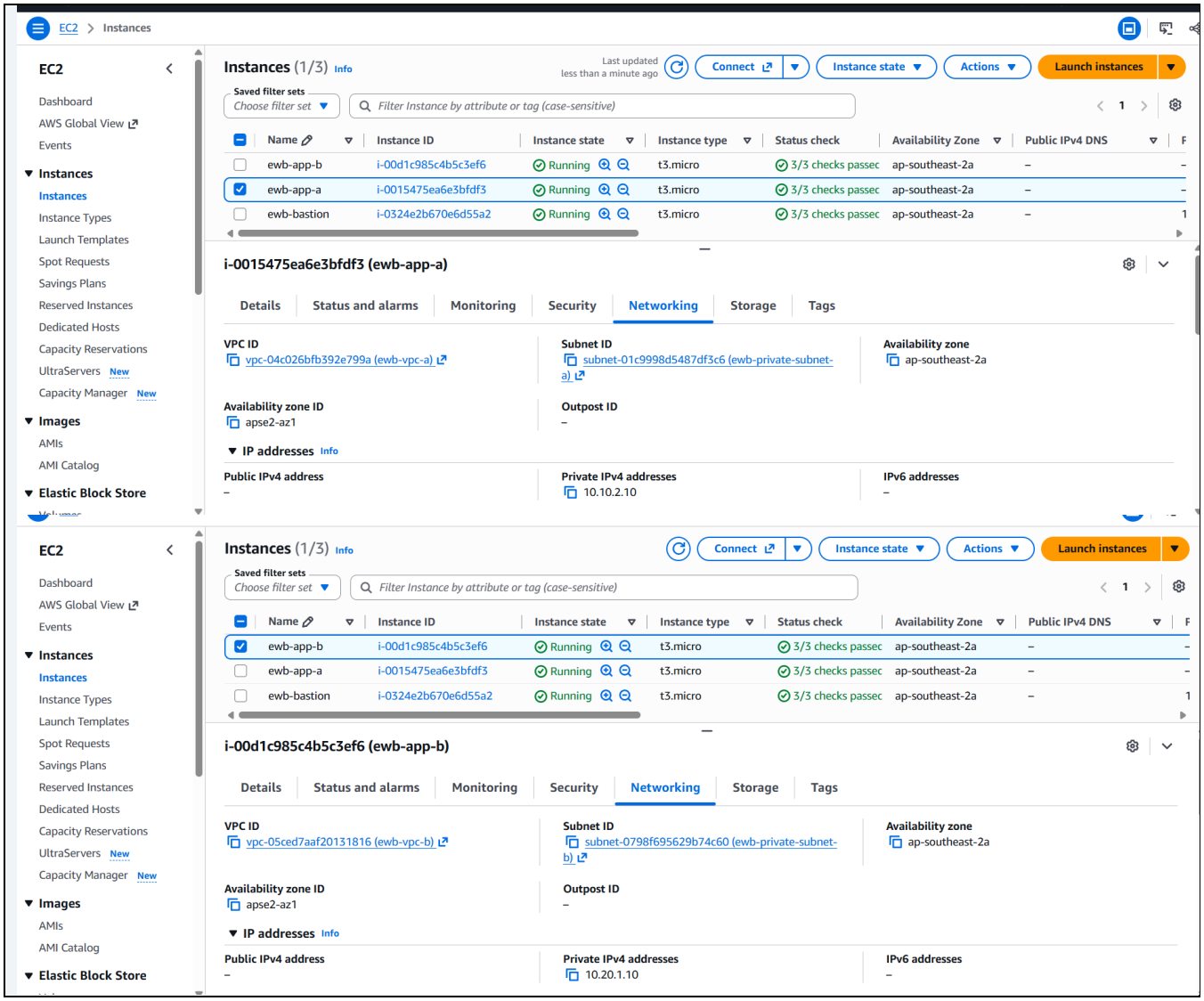


Figure 11 — Private instance networking



8. Connectivity Tests and Results

The tests use private destination IP addresses. The results below reflect tests reported during the lab. The baseline failure before peering was not captured at that time and is therefore marked as not evidenced.

Test	Command / method	Observed result	Outcome
Baseline before peering	App-A: ping -c 4 10.20.1.10	Not captured before peering was configured.	Not evidenced
Forward connectivity	App-A: ping -c 4 10.20.1.10	4 transmitted, 4 received, 0% loss.	Passed
Reverse connectivity	App-B: ping -c 4 10.10.2.10	4 transmitted, 4 received, 0% loss.	Passed
Private SSH	Laptop → Bastion → App-B 10.20.1.10	SSH shell established; hostname and hostname -I executed.	Passed
Controlled ICMP failure	Remove App-B inbound ICMP rule; ping from App-A	4 transmitted, 0 received, 100% loss.	Passed
ICMP recovery	Restore App-B inbound ICMP rule; repeat ping	4 transmitted, 4 received, 0% loss.	Passed

Figure 12 — App-A to App-B ping success

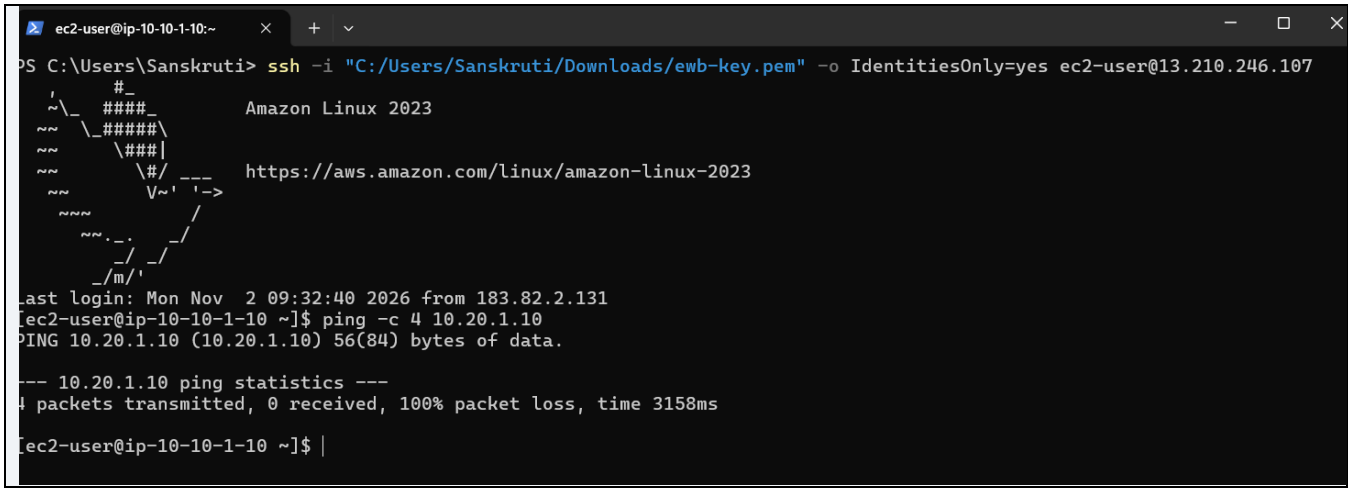


Figure 13 — App-B to App-A ping success

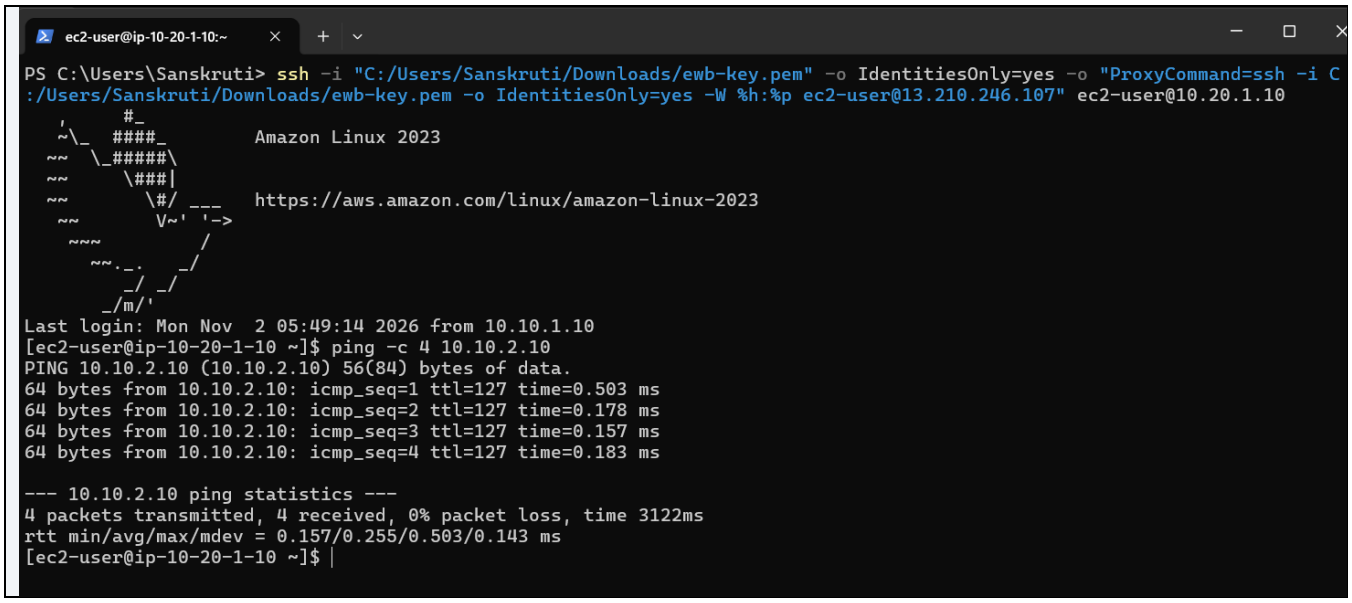


Figure 14 — App-B SSH proof

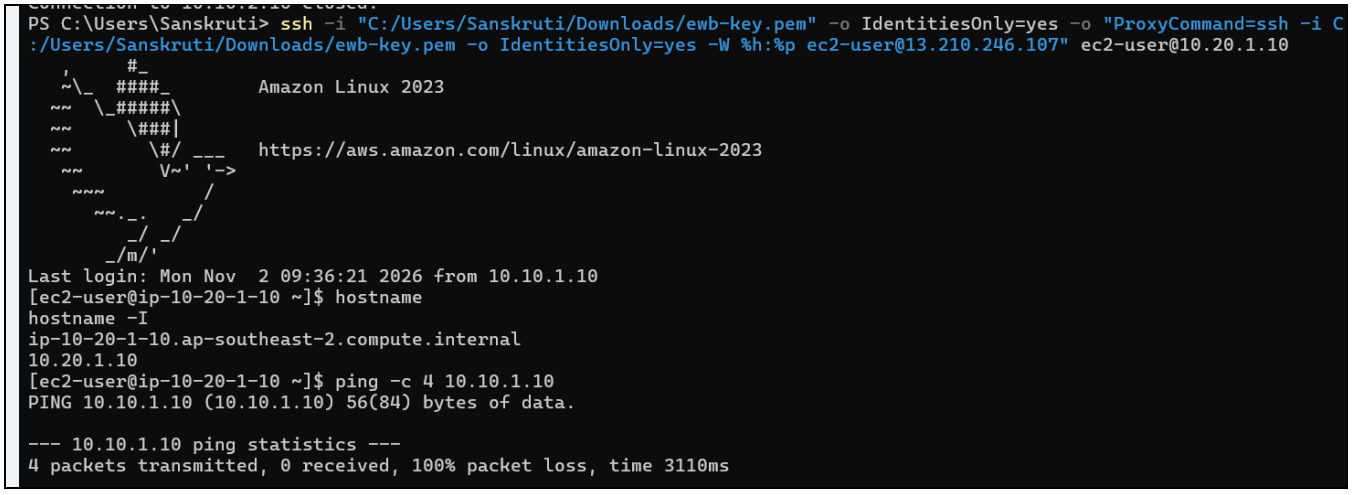


Figure 15 — ICMP rule removed / failure

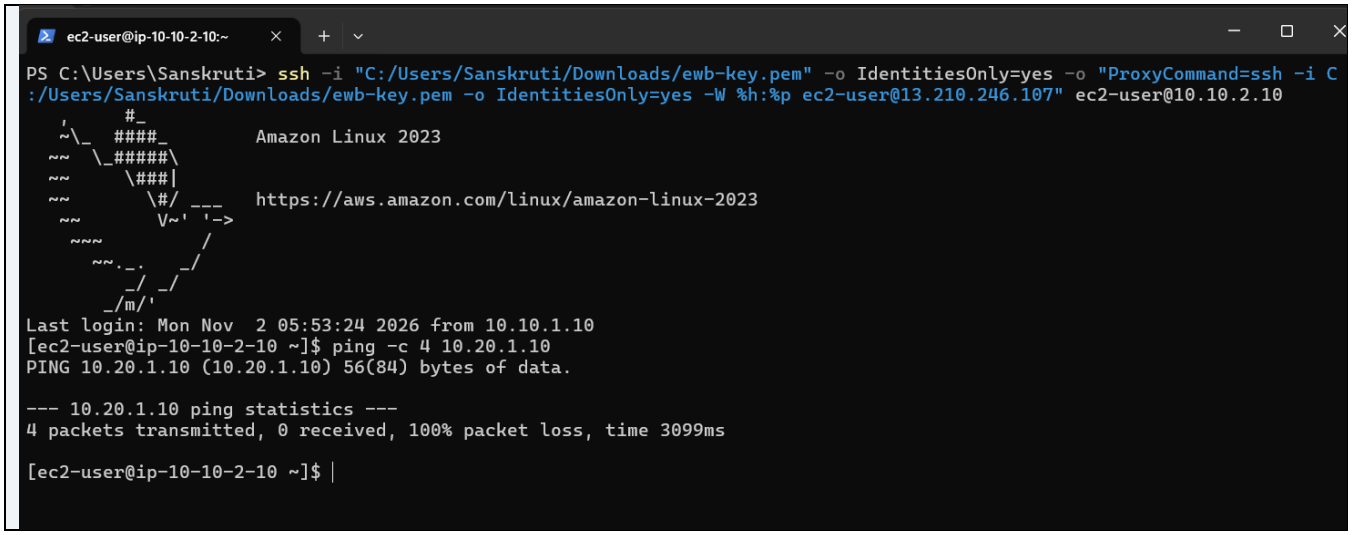
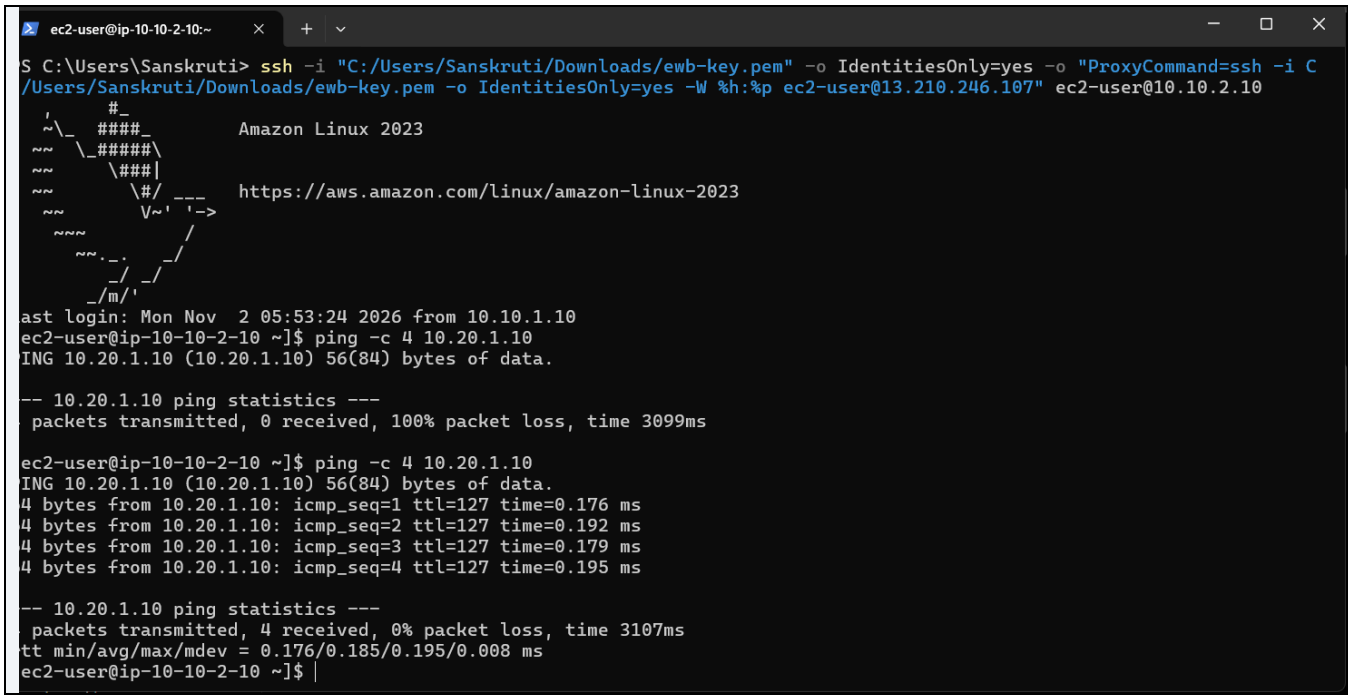


Figure 16 — ICMP rule restored / recovery



9. Traffic Flow Explanation

9.1 Forward path: App-A to App-B

App-A (10.10.2.10) sends an ICMP Echo Request to App-B (10.20.1.10). The VPC A private route table matches destination 10.20.0.0/16 and forwards the packet to ewb-a-b-peer. The peering connection delivers it to VPC B. App-B's security group must allow the required ICMP traffic from 10.10.2.10/32.

9.2 Return path: App-B to App-A

App-B sends the Echo Reply to 10.10.2.10. The VPC B private route table matches 10.10.0.0/16 and forwards the reply through the peering connection. The reply returns to VPC A and reaches App-A. A return route is essential for two-way communication.

9.3 SSH path to App-B

The administrator first connects to the bastion public IPv4 address. The bastion then forwards SSH traffic to App-B's private address 10.20.1.10. The VPC A public route table sends 10.20.0.0/16 through peering, and VPC B has the return route. App-B permits SSH from bastion private IP 10.10.1.10/32.

10. Controlled Failure and Recovery

The App-B inbound ICMP permission was temporarily removed. A new ping from App-A to 10.20.1.10 resulted in four transmitted packets and zero received packets (100% packet loss). This demonstrated that an active peering connection and valid routes do not bypass security-group filtering.

The ICMP permission was restored and the same ping was repeated from App-A. Four packets were received with 0% packet loss. This confirmed that restoring the inbound security rule restored connectivity.

11. Learning Outcomes

- VPC peering provides private communication between two VPCs with non-overlapping CIDRs.
- An Active peering connection alone does not guarantee reachability; forward and return routes are both required.
- Route tables must be associated with the correct subnets.
- Security groups control permitted inbound traffic at the instance level.
- A bastion host provides controlled SSH access to private instances.
- A controlled security-group change helps isolate connectivity problems.
- VPC peering is non-transitive.

12. Cleanup and Cost-Safety Checklist

Perform cleanup only after all evidence and the final report have been saved. Remove only resources created for this project.

☐ Terminate ewb-bastion, ewb-app-a and ewb-app-b.

☐ Confirm lab EBS volumes are deleted; remove leftover lab volumes or snapshots only after verifying ownership.

☐ Release any Elastic IP created for the lab.

☐ Remove the lab key pair if no longer needed; keep private keys out of this report.

☐ Delete peering routes and ewb-a-b-peer.

☐ After instance network interfaces disappear, delete the lab security groups.

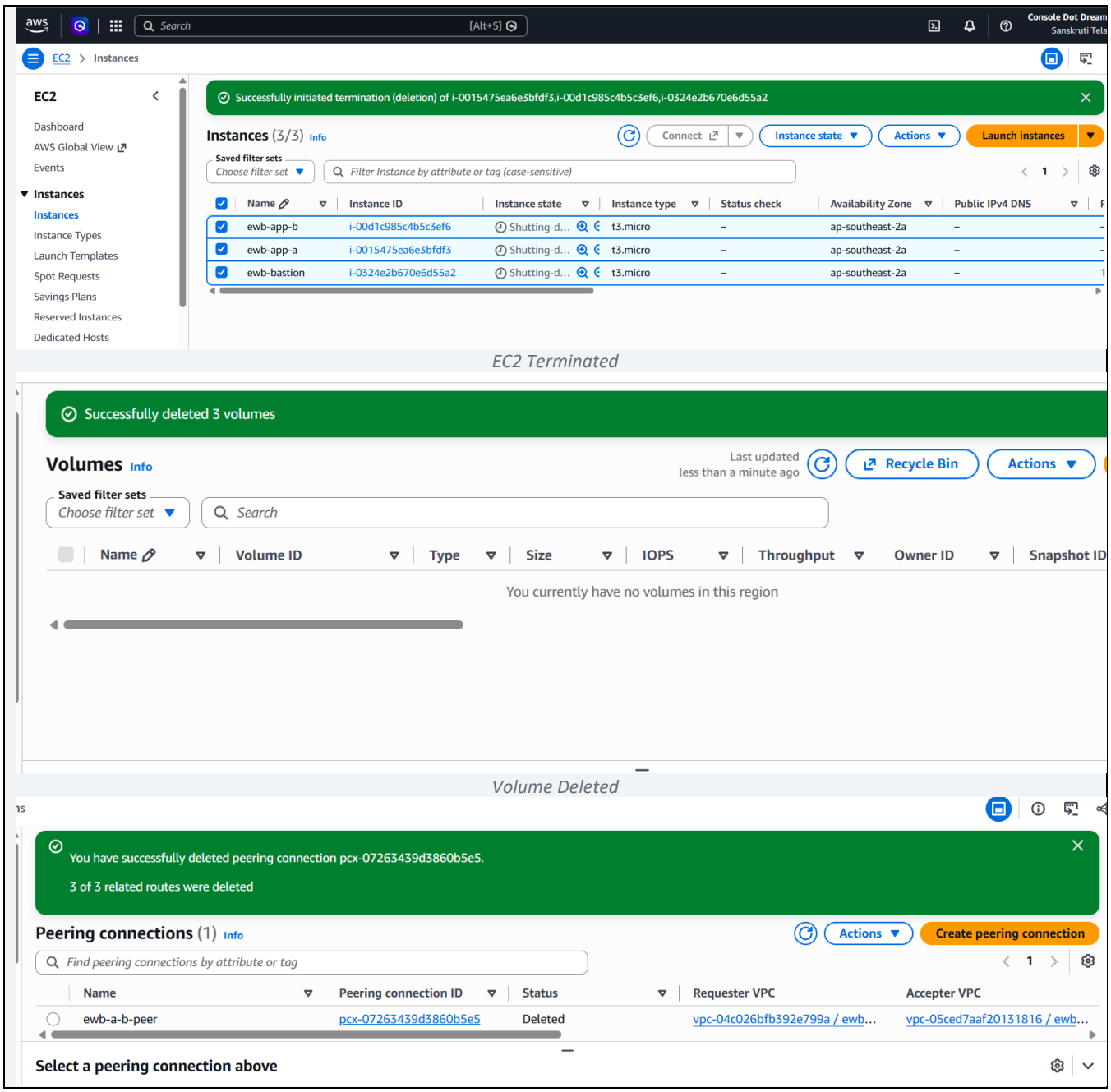
☐ Remove subnet associations as required, then delete lab subnets and custom route tables.

☐ Detach and delete ewb-igw-a, then delete both lab VPCs.

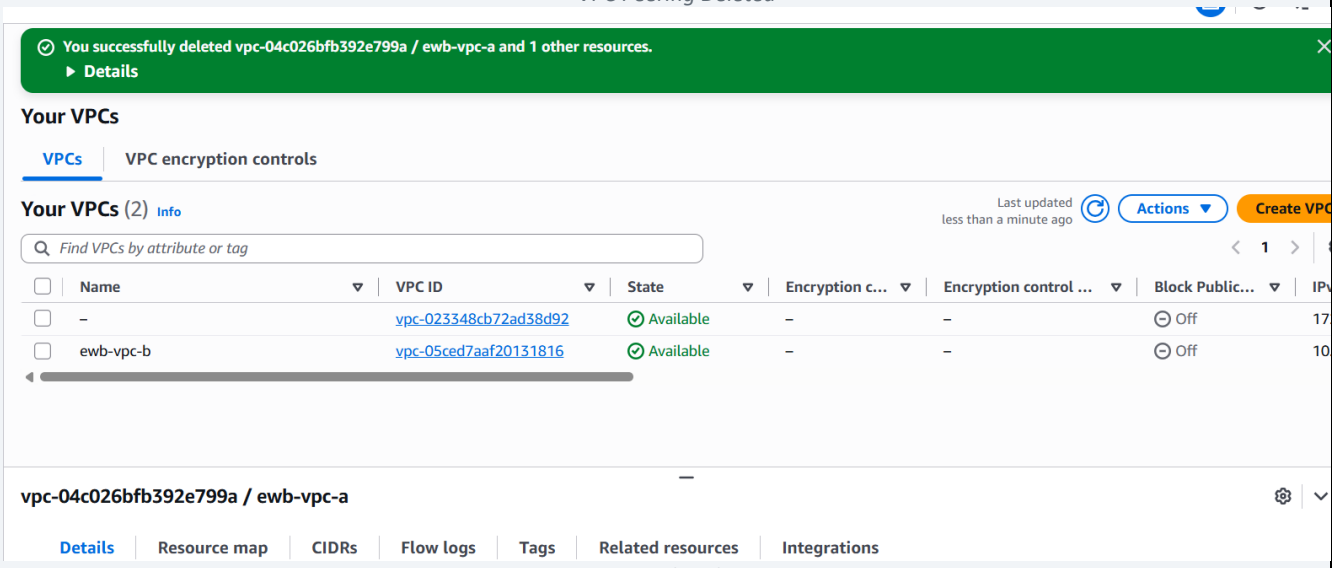
☐ Verify EC2, EBS volumes, public IPv4 allocations and VPC resources in ap-southeast-2.

☐ Check AWS Billing for remaining chargeable resources; usage reporting can lag.

Figure 17 — Cleanup evidence



VPC Peering Deleted



VPC Deleted

Cost note: EC2 compute, EBS storage and public IPv4 usage may incur charges. Stopping an instance does not necessarily stop EBS storage or public IPv4 charges. This project does not require a NAT Gateway.

13. Conclusion

The project demonstrated private communication between two VPCs using VPC peering. App-A and App-B exchanged ICMP traffic in both directions using private IPv4 addresses. App-B was accessed through the bastion without assigning it a public IPv4 address. The controlled ICMP test demonstrated how security-group rules can permit or block traffic independently of route configuration. The exercise reinforced CIDR planning, subnet associations, forward and return routes, least-privilege security rules and responsible AWS cleanup.

Appendix — Screenshot Index

Evidence	Suggested filename	Purpose
VPCs and subnets	VPC-A.png; VPC-B.png; Subnets.png	VPC names and CIDRs
Internet Gateway	Internet-Gateway-Attached.png	IGW attached to VPC A
Peering	VPC-Peering-Active.png	Active peering connection
Route tables	route-table screenshots	Routes and associations
Security groups	security-group screenshots	Inbound rules
EC2 and networking	EC2-Instances.png;networking screenshots	Instance details and no public IP
Ping tests	success screenshots	Forward and reverse reachability
SSH	App-B-SSH-Proof.png	SSH access to App-B
Failure / recovery	ICMP screenshots	Controlled block and restoration
Cleanup	Cleanup screenshot(s)	Resources removed and Region checked